

Day 10 Student Handout

Geometry Summer Credit | Right-Triangle Similarity and Weekly Review

Name:	Date:	Period:
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Learning Targets

I can use similarity in right-triangle altitude diagrams. I can review quadrilaterals, scale factor, and similar-triangle proportions before the weekly assessment.

Lesson 10A: Right-Triangle Similarity

Warm-Up

1.	Solve: $6/9 = x/24$.
2.	Triangle ABC is similar to triangle DEF. $AB = 8$, $DE = 20$, $BC = 12$. Find EF.
3.	A rectangle has diagonals $3x + 2$ and 29. Find x.

Concept Launch

Altitude to the Hypotenuse

When an altitude is drawn from the right angle to the hypotenuse, it creates two smaller right triangles. The small triangles are similar to the original triangle and to each other.

Problem	Work / notes
In right triangle ABC, altitude CD is drawn to hypotenuse AB. Name the three similar triangles: large triangle, left small triangle, and right small triangle.	Large: Left small: Right small: Similarity idea:
A right triangle has hypotenuse split into segments 4 and 9 by the altitude. The full hypotenuse is 13. What proportion can help find the altitude h?	Geometric-mean setup: $h^2 =$ $h =$
A right triangle has hypotenuse 25. One leg projects onto the hypotenuse in a segment of	$\text{Leg}^2 = \text{hypotenuse times projection}$: Equation:

length 9. Set up a proportion to find that leg.	
Why do right-triangle altitude problems rely on similarity?	Angles: Same shape: Proportions:

Useful Extension Formulas

Use When the Diagram Fits

If altitude h splits the hypotenuse into segments p and q , then $h^2 = pq$. If a leg has projection p on hypotenuse c , then $\text{leg}^2 = cp$. These come from similar triangles.

4.	If $p = 5$ and $q = 20$, find h using $h^2 = pq$.
5.	If $c = 18$ and $p = 8$, find the leg using $\text{leg}^2 = cp$.

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Lesson 10B | Weekly Review

Lesson 10B: Cumulative Review Before Assessment

Weekly Assessment Topics

Today's assessment reviews quadrilaterals, slope/distance/midpoint evidence, similarity, scale factor, proportions, and introductory right-triangle similarity.

Problem	Work / notes
Quadrilateral review: A parallelogram has consecutive angles $x + 40$ and $2x + 20$. Find x .	Equation: Answer:
Similarity review: A 5-foot person casts a 7-foot shadow. A tree casts a 35-foot shadow. Find the tree height.	Proportion: Answer:
Scale review: A rectangle is enlarged by scale factor 3. If the original perimeter is 22, what is the new perimeter?	Scale effect: Answer:
Coordinate review: Find the midpoint of A(-4, 8) and B(10, 2).	Formula: Answer:

Practice

6.	Solve: $8/12 = 14/x$.
7.	A rhombus has perimeter 68. Find each side length.
8.	A point (2, -5) is dilated by scale factor 4. Find the image point.
9.	In a right-triangle altitude diagram, $p = 6$ and $q = 24$. Find h .